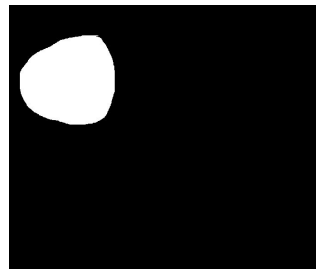
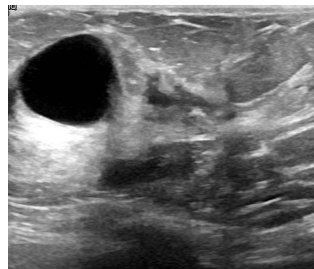


Towards Breast Cancer Tumor Segmentation

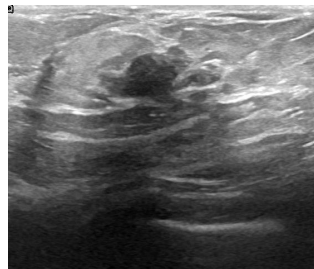
By: Joseph and Tenzin

Introduction

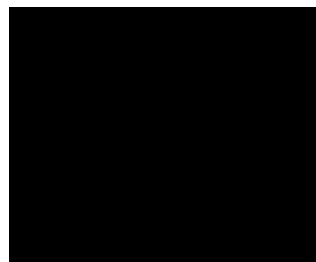
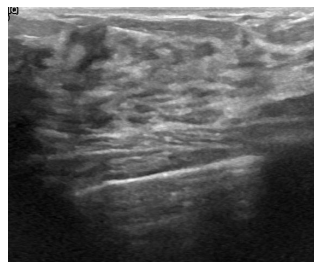
- Comparing variations of U-Net architecture for breast tumor segmentation
 - [Dataset](#): Ultrasound breast tumor cancer images
 - Benign (891 images)
 - Malignant (421 images)
 - No Tumor (266 images)
 - Images from women ages 25 to 75
 - Collected in 2018 (Al Dhabyani et al.)
- Purpose:
 - Early detection
 - Feature extraction
 - Eventual classification



Benign



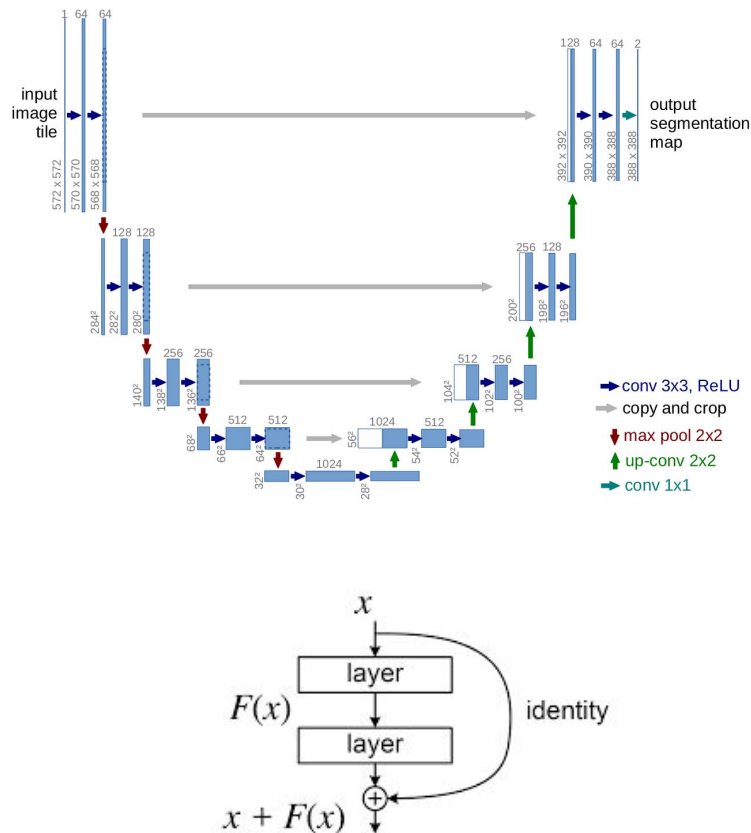
Malignant



No tumor

Methodology

- Baseline U-Net (Ronneberger et al.)
 - Fine tuning
- Data Augmentation
 - Flip/Rotation
- Loss Functions
 - Binary Cross Entropy
 - Dice/IoU Loss Functions
- Residual Connections (Zhang et al.)
- Attention (Oktay et al.)



Evaluation

- 80/20 Training/Test Split
 - Cross Validation w/ training dataset for hyperparameter fine-tuning
 - Stratified based on benign, malignant, normal cases
- Evaluation Metrics
 - Intersection over Union
 - Dice
 - Minimize FN
 - Precision
 - Minimize FP
 - F1-Score

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$


$$\text{Dice} = \frac{2 \times \text{Area of overlap}}{\text{Total area}} = \frac{2 \times \text{Prediction} \cap \text{Ground truth}}{\text{Prediction} \cup \text{Ground truth}}$$
